

# Sterilization of *Amyelois transitella* (Lepidoptera: Pyralidae) by electron beam technology

Running Title:

eBeam Sterilization of Navel Orangeworm

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## Abstract

### Background

Use of the sterile insect technique (SIT) for control of arthropods relies on the mass release of sterile individuals to suppress wild populations of the target pest. Insects are commonly sterilized with gamma radiation from either cobalt or cesium source material, which poses some human health and environmental risks. One potentially safer alternative could be the use of electron beam radiation. In this study, we evaluated whether a commercial electron beam system could serve as a practical non-nuclear alternative for sterilizing *Amyelois transitella* (Lepidoptera: Pyralidae), a major pest of tree nuts in California. To do this, we tested the effects of different doses of electron radiation (150-550 Gy) on longevity, flight ability, fecundity and egg viability of *A. transitella*.

## Results

We were able to completely sterilize female *A. transitella* at 177 Gy and males at 309 Gy without any significant impacts on their longevity or flight ability, demonstrating for the first time that electron radiation can reliably sterilize this pest.

## Conclusion

Our research shows that electron radiation can be effective in sterilizing insects without causing significant harm to their health and fitness, and in this way may provide a viable non-nuclear alternative with the potential for high throughput sterilization in an SIT program.

## **Keywords**

Sterile insect technique, activity, longevity, mating, fertility, fecundity

# 1 Introduction

In arthropod pest management, the sterile insect technique (SIT) relies on the introduction of large numbers of mass-reared and sterilized individuals to suppress wild populations of a target pest, and has been used worldwide for pest management over the past 75 years<sup>1</sup>. For instance, SIT has been used successfully against agricultural pests such as the pink bollworm (Lepidoptera: Gelechiidae: *Pectinophora gossypiella*), livestock pests such as the new world screwworm (Diptera: Calliphoridae: *Cochliomyia hominivorax*), and disease vectors such as the yellow fever mosquito (Diptera: Culicidae: *Aedes aegypti*)<sup>2–7</sup>. To date, most SIT programs have relied upon the use of ionizing radiation, most commonly gamma radiation, to sterilize insects<sup>5,8–10</sup>.

Ionizing radiation for SIT programs is usually produced by gamma irradiators using a cobalt-60 or cesium-137 source material<sup>5,8,10,11</sup>. These sources are long-lasting and reliable but also come with a wide range of human health and environmental safety risks, as well as high replacement

costs<sup>12</sup>. As such, there has been growing interest to find viable non-nuclear alternatives to gamma radiation for sterilization of insects, such as electron beam technology<sup>8,13–15</sup>.

An electron beam is a stream of accelerated free electrons that are focused on a specific area. In this form, electrons can damage DNA. This is achieved either directly, when electrons hit a DNA molecule, causing it to break apart, or by creating reactive oxygen species when reacting with water molecules in the vicinity of DNA<sup>16,17</sup>. Electron beams can be used to sterilize various materials, surfaces and substances, such as surgical equipment, medicine, produce and even water<sup>10,12</sup>.

Navel orangeworm (*Amyelois transitella*, Lepidoptera: Pyralidae) is an economically important pest of almonds, pistachios, and walnuts in California<sup>18,19</sup>. These crops are grown on approximately 800,000 ha in California and represent more than 15% of the annual farm-gate value of all of California agriculture<sup>20</sup>. *Amyelois transitella* threatens this value through direct damage to nut kernels and indirectly by contamination with *Aspergillus* fungi that produce carcinogenic aflatoxins<sup>18</sup>. The pest potential of *A. transitella* comes from several life history characteristics. It is multivoltine<sup>21,22</sup>, a polyphagous feeder on overripe, damaged, cracked, or mummified fruits or nuts<sup>23</sup>, and capable of inter-orchard dispersal<sup>24</sup>. Best management practices include crop sanitation<sup>18,25</sup>, mating disruption<sup>26–28</sup>, monitoring<sup>29</sup>, targeted chemical control<sup>28,30</sup>, and timely harvest<sup>19,31–33</sup>. The landscape and life history characteristics of *A. transitella* have led to interest in area-wide approaches<sup>34</sup>, possibly including the use of sterile insect technique<sup>11</sup>.

Electron beam (eBeam) irradiation has been used for sterilization of *Ceratitis capitata* (Diptera: Tephritidae)<sup>5,8</sup> and considered for use with *Cochliomyia hominivorax* (Diptera: Calliphoridae)<sup>13,35</sup>. A dose of 95 Gy was reported for *C. capitata*<sup>5</sup>, and a dose of 40–70 Gy was modelled for *C. hominivorax*<sup>13</sup>. However, we are not aware of published studies of dosimetry for insect sterility using eBeam, as have been reported for various species using gamma and X-ray<sup>8,36,37</sup>.

Generally, Lepidoptera are more radiotolerant than other orders, and Pyralidae tend to be among the more radiotolerant moth families<sup>8,38</sup>. Previously gamma and X-ray radiation have been used with *A. transitella* for sterile insect work<sup>5,36,39–41</sup>. An earlier study with gamma radiation generated by cobalt-60 at a rate of approximately 33 Gy/min found that 500 Gy was necessary for complete sterility of both sexes when applied to mature pupae<sup>36</sup>. Currently, complete sterility is obtained in both sexes with 300 Gy gamma generated by cobalt-60 at a rate of approximately 70 Gy/min<sup>5,40,41</sup>. A study using X-ray found complete sterility in both sexes of *A. transitella* at 125 Gy<sup>39</sup>. It should be noted, however, the X-ray irradiator used had an input voltage of  $\leq 100$  kEV, were unshielded and a dose rate of approximately 55 Gy/min<sup>39</sup>, and is therefore very different from commercial X-ray irradiators more commonly used, which typically have an input voltage of 160 kEV and a dose rate of around 10 Gy/min<sup>42,43</sup>. Studies using this type of equipment suggest a sterilizing dose similar to that found in the gamma studies (authors, unpublished data).

Here, we evaluated whether electron beam irradiation can reliably sterilize *A. transitella* without compromising traits important for a successful SIT program, such as insect longevity, flight ability and mating performance.

## 2 Methods

### 2.1 Navel Orangeworm (*Amyelois transitella*) Colonies

All *Amyelois transitella* (Lepidoptera: Pyralidae) moths were reared at the USDA Agricultural Research Service (ARS) San Joaquin Valley Agricultural Sciences Center (SJVASC) in Parlier, CA. Moths were reared at 26°C under a 14:10 L:D photoperiod with 60% relative humidity. Larvae

were fed a standard wheat bran diet throughout their development <sup>44</sup>. Unmated adult moths, separated by sex, were isolated within 36 h of eclosion for use in the irradiation assays.

## 2.2 Preparation of Moths for Irradiation and Dosimetry

Prior to irradiation, individual moths were arranged within a 127 × 82.55 mm 48-well plate (6 × 8 wells) (Corning®Costar®HTS Transwell®, CLS 4395, Corning, NY). To ensure even spread, optimal dose distribution and prolonged cooling, moths were placed individually into well plate cells, with 5 moths per row, leaving one empty well in each row, in total 30 individuals per well plate in 10 plates (Supplemental Figure S1). To measure the applied radiation dose, an alanine tablet (FWT-50-30, DE582, Harwell/Steris, Didcot, Oxfordshire, UK) was placed into the empty well in every other row. We included two controls in the experiment. One control group was kept under colony conditions at the USDA-ARS San Joaquin Valley Agricultural Sciences Center, while the other control group was placed into the 48-well plates as described, and traveled with the other moths to the eBeam facility and back, but without any irradiation.

## 2.3 Transportation of Moths to the Commercial eBeam Facility

Following placement into the well plates, moths were cooled to  $8.0 \pm 1.0$  °C to reduce movement and stress. The plates were then wrapped in multiple layers of cushioned foil (Bubblewrap®/Sealed Air Corporation, Charlotte, NC) and placed in an electric cooler (30l portable refrigerator, Euhomy, Sacramento, CA) set to 8 °C that included a temperature sensor (HOBO Temperature/RH Data Logger, MX2301A, Onset Computer Corporation, Bourne, MA) to confirm temperature during transport. After packaging, the moths were driven from the Kearney Agricultural Research and Extension Center (Parlier, CA) to the commercial eBeam facility (Steri-Tek, Fremont, CA), which was about a 3-hour drive away. Moths were then held in the cooler

overnight until irradiation the following morning. Such trips were made on two separate occasions for the fitness tests, and another two such occasions for the fertility tests (Table 1). These replicates in time are referred to as “Runs” in this paper.

## 2.4 Irradiation Procedures

At the eBeam facility, each well-plate was randomly assigned to one of the intended dose treatments (0, 200, 300, 400 and 500 Gy), then placed into Styrofoam racks ( $26.67 \times 19.05 \times 2.54$  cm) that were then inserted into cardboard sleeves ( $31.75 \times 24.13 \times 3$  cm) so they could be secured within the conveyor frames (Supplemental Figure S2) that moved through the electron beam (Information about eBeam device according to CIRPER <sup>45</sup> Supplemental Table S3). All sleeves were processed identically, and the control sleeve underwent the same handling procedure except for irradiation. During irradiation, the frames passed through the electron beam on a conveyor belt. The beam delivered a constant dose rate of 166 Gy/min, and for each treatment the applied dose was adjusted by changing the conveyor speed.

Following irradiation, all well-plates were removed, repackaged as before, and returned to the portable cooler. The full irradiation procedure in the commercial facility took approximately 4 hours, followed by 3 hours of driving back to the Kearney Agricultural Research and Extension Center. Upon arrival at the laboratory, moths were assigned to either (i) flight ability and longevity assays or (ii) mating and sterility assays in alternating experimental runs because the number of *A. transitella* available per run (300–360 individuals) was insufficient to conduct all assays simultaneously. Two runs were completed for each assay, with a total of 120 individuals (60 females and 60 males) evaluated per treatment and assay. Each run consisted of 30 individuals of each sex per treatment.

For flight ability and longevity assays, groups of 10 same-sex *A. transitella* were placed in a flight cage. Each flight cage constituted one replicate, resulting in three replicates (30 individuals) per treatment and sex per run.

For mating and sterility assays, 30 treated individuals of each sex and treatment were individually paired with untreated partners in each run. Each mating pair constituted a single replicate, resulting in 30 replicates per treatment and sex per run.

## 2.5 eBeam Dosimetry

As mentioned, dosimetry of the eBeam system was determined using alanine tablets irradiated alongside the moths in the same well-plates. The tablets were purchased from and read by the National Center for Electron Beam Research at Texas A&M University (College Station, TX) to determine absorbed dose. Tablets were shipped to Texas A&M University one day after irradiation. Alanine tablets were included in all well-plates brought to the eBeam facility, including the unirradiated control (i.e. Trip-Control treatment), but not in the control well-plates that remained in the laboratory (i.e. Lab-Control treatment).

## 2.6 Fitness Assay (Flight Ability and Longevity)

Adults from two trips to the eBeam facility were tested for flight and longevity (Table 1). Flight ability was evaluated using flight cylinders, a well-established assay for insect quality <sup>46</sup>. Flight cylinders consisted of polyvinyl chloride (PVC) pipe (diameter = 16 cm, height = 16 cm) with interior surfaces coated with fluon (Insect-a-Slip™, Bioquip, Rancho Dominguez, CA) to ensure that moths escaped only by flight. Individual cylinders were placed within insect cages (30 × 30 × 30 cm; BugDorm-4E3030 Insect Rearing Cage, Bugdorm, Taiwan) and held under a summer

photoperiod (14:10 L:D) at ambient temperature ( $26 \pm 3^{\circ}\text{C}$ , 60% RH). Moths were placed in cylinders in groups of 10, separated by sex and treatment. The number of moths inside and outside the flight cylinders was recorded after 24 h, after which observations continued for the longevity assay.

For the longevity assay, moths were kept under the same conditions as in the flight assay (summer photoperiod 14:10 L:D; ambient temperature  $26 \pm 3^{\circ}\text{C}$ , 60% RH). The flight cylinders were removed, and water-soaked cotton balls (Jumbo Cotton Balls, Equate, Brandonville, AR) were added to all cages. These cotton balls were rehydrated every two days for the duration of the experiment. The number of surviving moths was noted daily. The experiment was terminated after 14 days.

## 2.7 Fertility Assay (Mating, Fecundity, and Larval Development)

Adults from the other two trips to the eBeam facility (Table 1) were tested for reproductive ability. Individual moths were caged with a fertile, unmated moth of the opposite sex in plastic 30 mL portion cups and allowed to mate over a 24-hour period. The female individual of each pairing was then transferred to a 473 ml glass jar for the oviposition assay. The jars were covered with a round filter paper held in place with a banded lid and then placed on their sides so that the filter paper served as a vertical oviposition substrate.

Previous work has shown that after four days over 75% of eggs have been deposited<sup>47</sup>. In addition, *A. transitella* is known to cannibalize other larvae<sup>48</sup>, so if the first larvae emerge and no diet is present, they could start feeding on each other, changing the results obtained. Therefore, after three days or at time of death female moths were retained individually pending examination for mating status and the oviposition paper was removed and the eggs counted. The paper was then placed egg-side down on wheat bran diet in a petri dish. The *A. transitella* eggs were allowed to incubate for 25 days at  $26^{\circ}\text{C}$ , 60 % RH and a 14:10 h (L:D) photoperiod.

To determine mating status, the bursa copulatrix of all female moths was dissected to confirm that a single spermatophore was present. To determine fecundity, all eggs deposited onto the filter paper were recorded. Finally, to determine larval development success, the number of pupae per petri dish was noted after the 25-day incubation period.

## 2.8 Data Analysis

All data was analyzed using the R Statistical Program, version 4.5.2<sup>49</sup>. The packages used for the various analyses included glmmTMB<sup>50</sup> to fit and compare generalized linear models, DHARMA<sup>51</sup> to examine model fit, and car<sup>52</sup> to conduct type II analysis of deviance using likelihood ratio or Wald chi-square tests. Generalized linear models (GLMs) were used to analyze the data (Supplemental Table S4). Response variables including the proportion of moths escaping from the flight cylinder, the proportion of moths that survived >14 days, the proportion of moths mated, the number of eggs laid, and the number of F1 that developed to pupa with dose, sex, run, and all two-way interactions as predictors. For the flight cylinder and longevity analyses, the cage containing 10 adults was the experimental unit. A quasibinomial distribution was used with the GLM, and dose (Gy) was modeled as a continuous variable. In addition, the data were plotted with linear regression lines including the 95% confidence interval of the mean in order to illustrate trends.

For analyses of fertility assays for the other two runs (with mating, fecundity, and pupal count as response variables) the experimental unit was individual adults. The proportion of adults that mated was modeled with dose as a continuous variable, using a GLM with a binomial distribution. However, for egg and pupa counts, doses were analyzed as a categorical variable in order to better fit models to data with overdispersed frequency and complete separation. Doses of 176 and 177 Gy were both categorized and described as “175 Gy”, doses of 208 and 211 Gy were

categorized and described as “210 Gy”, and doses of 405 and 445 Gy were categorized and described as “420 Gy”.

For analysis of the oviposition and larval development assays, we only used data from mated individuals and rejected all data gathered from individuals who were found to have not mated. Oviposition data were analyzed using a GLM with a negative binomial distribution, and counts of F1 that developed to the pupal stage were compared using a GLM with a zero-inflated negative binomial distribution. Based on findings from the GLMs, counts were compared between doses separately within each level of sex  $\times$  run. The non-parametric Kruskal-Wallis test was used to determine if there were significant differences between levels of dose, and the Dunn test with the Holm adjustment for multiple comparisons was used to compare between doses.

## 3 Results

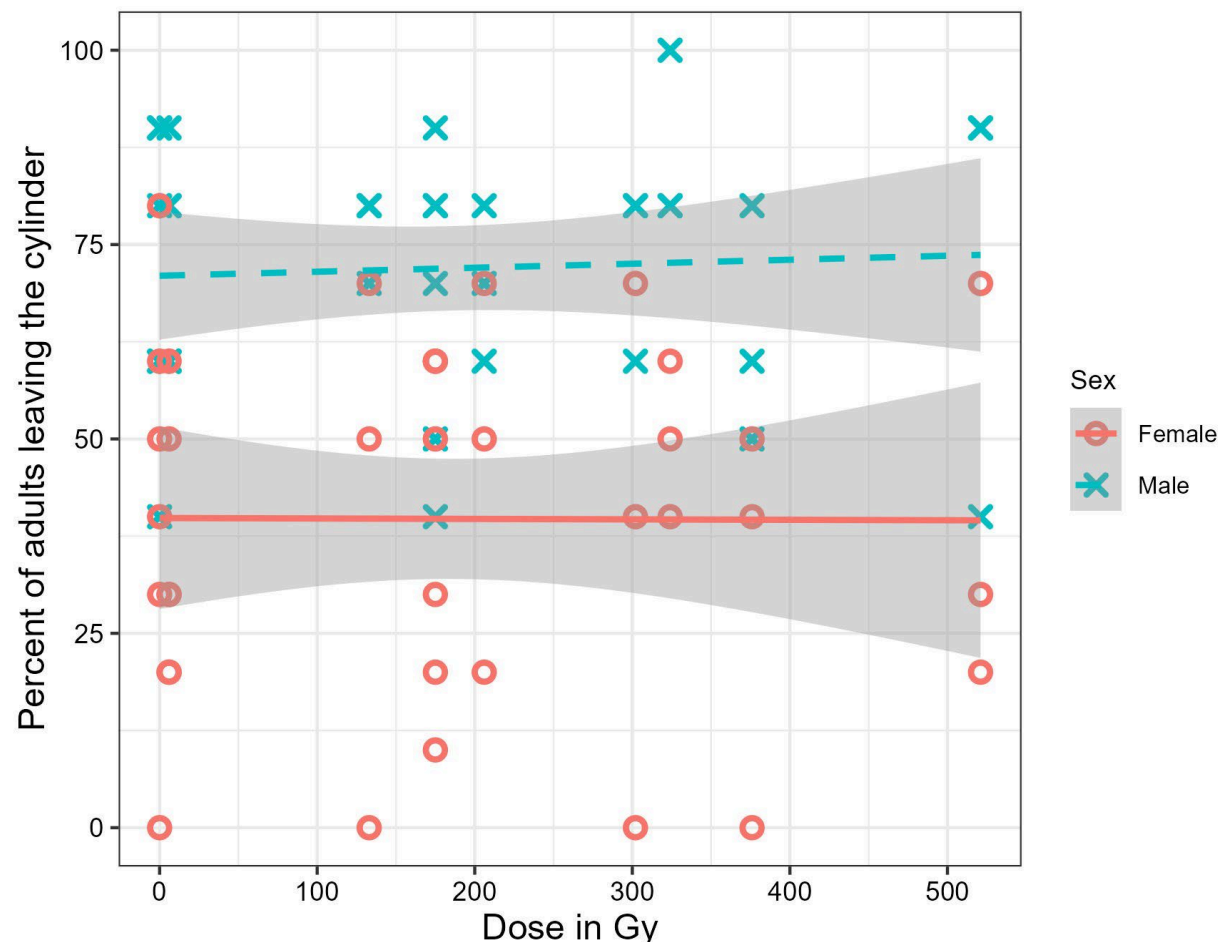
### 3.1 Irradiation and Dosimetry

As mentioned, conveyor belt speed was adjusted to achieve the desired irradiation treatments of 100, 200, 300, 400 and 500 Gy. Readings from the alanine tablets indicated that the applied doses were close to, but not exactly, at the levels desired. A comparison of the intended and applied treatment doses are provided in Table 1. Regardless, a low-to-high range of electron beam radiation doses were successfully applied to the moths.

### 3.2 Flight Ability and Longevity Assays

#### 3.2.1 Flight Ability Assay

Flight ability was not affected by increasing electron beam dose (Table 2), in either female or male *A. transitella* (Fig. 1). Across all doses, males escaped from flight cylinders more often than females (males  $7.19 \pm 0.26$  out of 10, females  $3.97 \pm 0.37$  out of 10;  $t(62.8) = 7.031$ ,  $P < 0.001$ ).

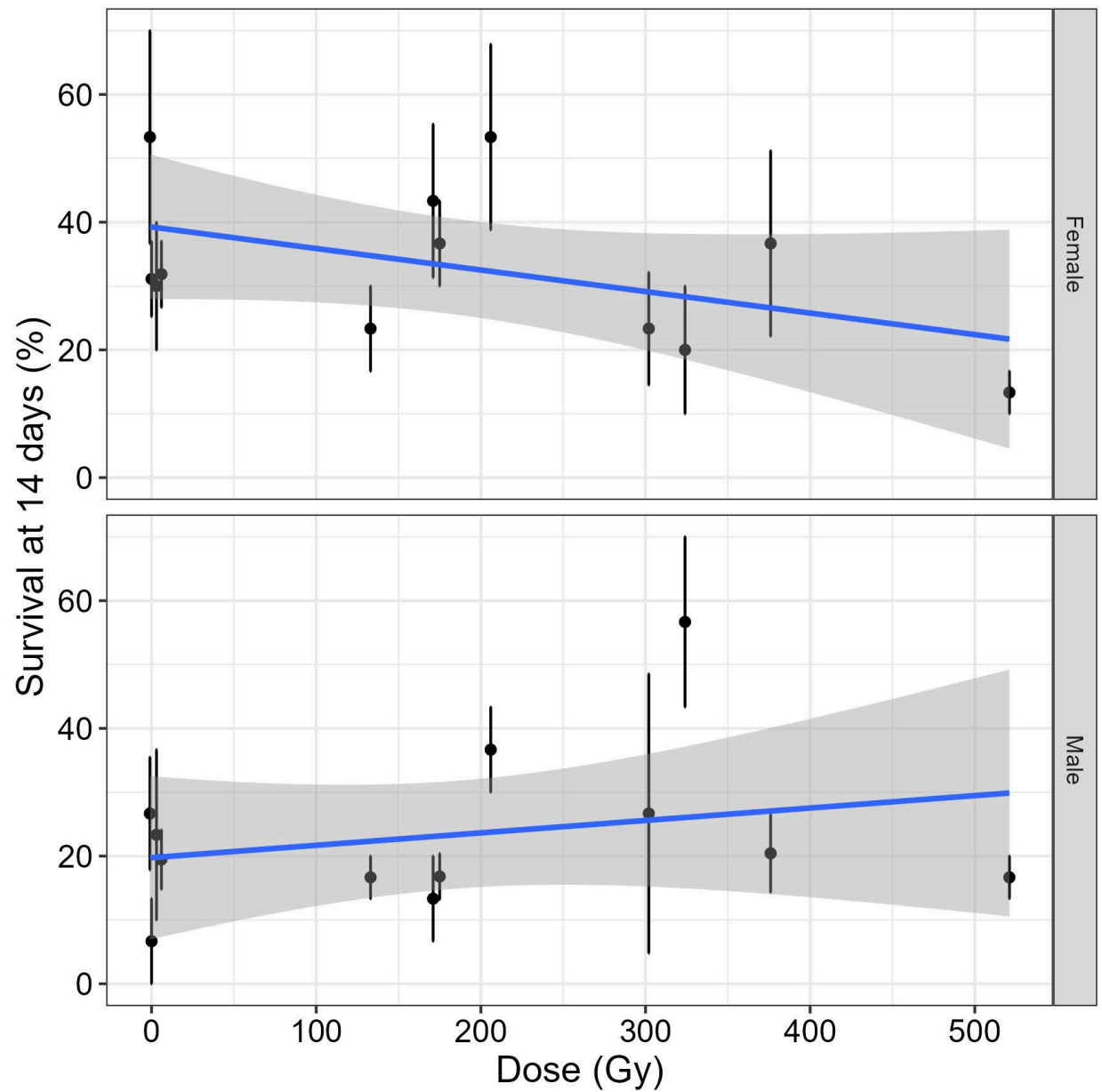


**Figure 1** Effect of eBeam irradiation indicated by dose (Gy) on flight ability of female (red rings) and male (cyan x's) *A. transitella*. Symbols indicate individual flight cylinder experiments (groups of 10 moths). The lines indicate the linear regression fit for each sex, and the grey area indicated the 95% prediction interval.

### 3.2.2 Survivorship Assay

Radiation dose did not have a significant effect on longevity (Table 3). While there was no significant effect of dose on longevity, the dose \* sex interaction was significant, and the effect of

run (cohort) was marginally significant (Table 3, Fig. 2). More females than males lived longer than 14 d ( $33 \pm 3.2\%$  vs  $23 \pm 3.1\%$ ).

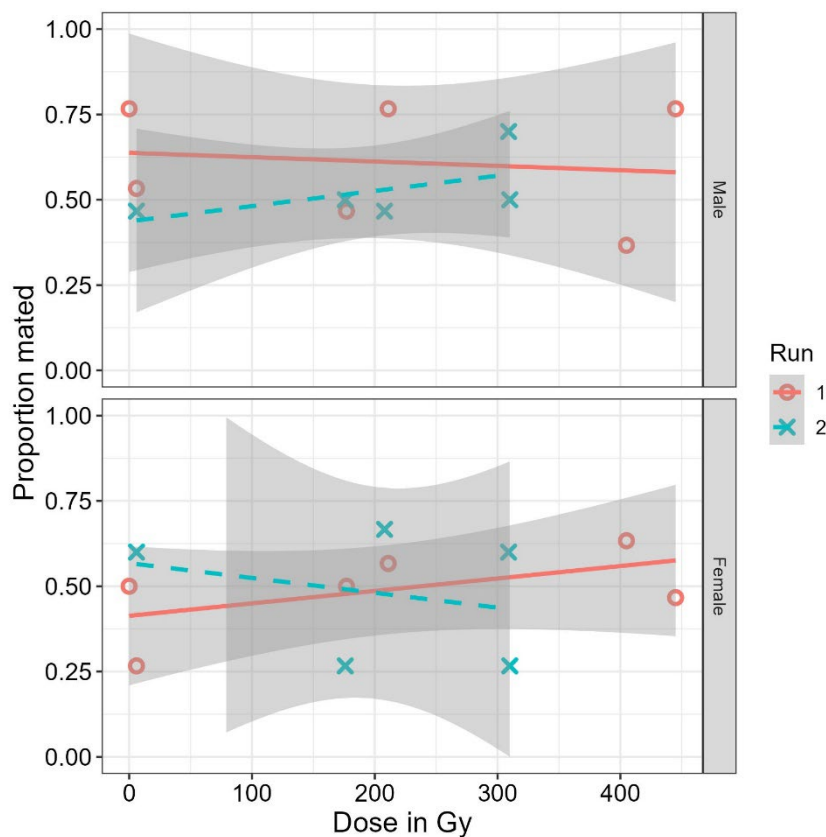


**Figure 2** Effect of eBeam irradiation indicated by dose (Gy) on the proportion of *A. transitella* that survived >14 d. The lines indicate the linear regression fit for each sex, and the grey area indicates the 95% prediction interval.

### 3.3 Mating, Oviposition, and Development Assays

#### 3.3.1 Mating Assay

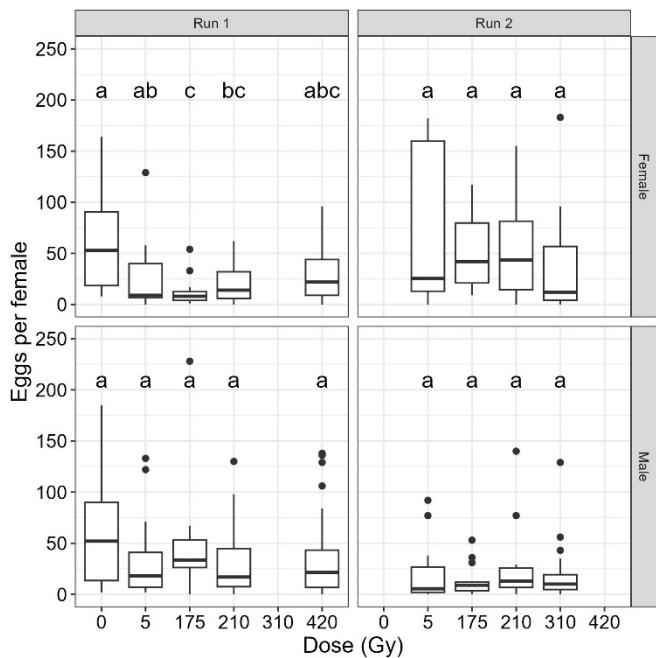
The effect of sex on the proportion of moths that mated was marginally significant (Table 4). Other effects, including dose, were not significant (Table 4, Figure 3).



**Figure 3** Effect of eBeam irradiation indicated by dose (Gy) on the mating ability of female and male *A. transitella* indicated by the proportion of mated individuals after two days in the first (red rings) and second (cyan x's) runs with an untreated individual of the opposite sex out of groups of ten. Linear regression and the 95% confidence interval for each level of sex  $\times$  run are respectively indicated by the straight lines and shaded areas.

### 3.3.2 Oviposition Assay

There were significant effects on total oviposition per female due to dose, and there was also a significant interaction of sex \* run (Table 5). Within sex \* run, differences between doses tested were only significant for females in Run 1 (Fig. 4). Eggs were produced from pairings for over 90% of mated adults tested, and within sex \* run there was no significant difference in eggs per female between the lowest and highest doses.

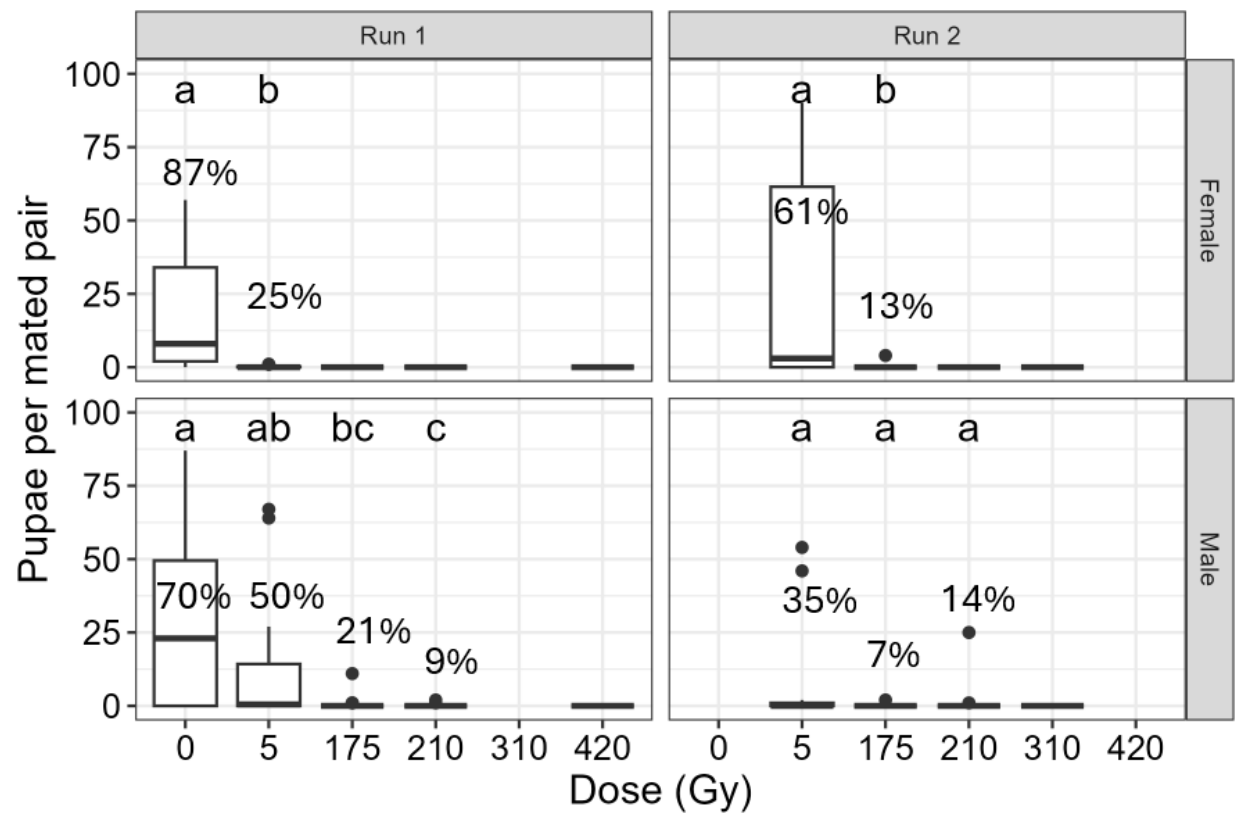


**Figure 4** Box plots illustrating the effect of eBeam irradiation on the fecundity of *A. transitella*. Sex indicates the sex irradiated; for “Male” this means eggs per non-irradiated female paired with an irradiated male. Treatment groups in the same panel with different letters are significantly different. The 0 treatment indicates moths that were not taken to the eBeam facility, whereas the 5 Gy treatment indicates moths that were taken to the eBeam facility but not exposed.

### 3.3.3 Development Assay

The GLM examining effects on the count of pupae produced by experimental pairings revealed highly significant effects due to dose and the sex \* run interaction, in addition to significant effects

of run and dose \* run and a marginally significant effect due to dose \* sex. When treatment differences were compared within sex \* run, dose dependent effects were evident (Fig. 5). This dose-dependence was generally more incremental when displayed as percent fertile (i.e., percent of pairings that produced pupae) (Fig. 5). The lowest doses resulting in no pupae were 210 Gy for females (0 out of 37 pairs), and 310 Gy for males (0 out of 36 pairs).



**Figure 5** Box plots indicating pupae per female following successful mating. Treatment groups in the same panel with different letters are significantly different (treatments with 0 pupae were not included in this analysis). The numerals indicate the percent of pairings that produced >0 pupae.

## 4 Discussion

Our findings demonstrate that electron beam irradiation of *A. transitella* does not adversely affect flight ability, longevity, mating success, or fecundity. Although some significant differences among individual dose treatments were observed, no dose-dependent trends were evident for any of these parameters. In contrast, fertility declined consistently with increasing electron beam dose. Complete sterility was achieved at doses >200 Gy for females and >300 Gy for males, without compromising other fitness traits. The greater radiotolerance of males compared to females has been documented across numerous insect species<sup>53–55</sup>. Since no detrimental effects beyond reduced fertility were observed with females who received >300 Gy, it may be possible to sterilize both males and females simultaneously using the higher dose required for male sterility. Doses below the sterilizing threshold still reduced fertility substantially, although a small number of offspring did successfully develop. Similar results have been reported for X-ray irradiation of *A. transitella* before, where fertility reductions exceeded 99%, yet complete sterility required considerably higher doses<sup>39</sup>. Individual variation in radiotolerance or DNA repair capacity may explain the survival of these rare offspring<sup>56–58</sup>. Alternatively, these observations may reflect stochastic variation in radiation-induced damage<sup>58</sup>. Given the well-documented phenomenon of inherited sterility in Lepidoptera, it is possible that surviving offspring would themselves exhibit reduced fertility or even be fully infertile<sup>59–61</sup>. Unfortunately, F1 sterility could not be evaluated here.

While electron beam irradiation proved highly effective for sterilizing *A. transitella*, we also found that handling, transportation, and associated stress appeared to reduce fertility in untreated controls brought to the commercial facility, compared with controls that remained in the laboratory. While different methods of rearing and handling *A. transitella* can affect their performance<sup>40,41</sup>, this is a separate issue from determination of the sterilizing dose, which was our goal here. Still,

as with any SIT program, the rearing, handling, sterilization, transport and release procedures for *A. transitella* will all need to be optimized to develop an areawide program. Here, as a first step, we demonstrated the potential of electron beam irradiation as a high-throughput method of sterilization for *A. transitella* with minimal impacts on performance.

The present study indicates that a 10 MeV eBeam can, in principle, rapidly and effectively sterilize *A. transitella* while retaining adult flight capacity and longevity. The eBeam unit used in this study was capable of delivering sterilizing doses (>300 Gy) within approximately 2 min. With an alternative setup, large quantities of insects could be continuously transported through the electron beam using a conveyor system, enabling high-throughput irradiation. Currently eBeam is not widely used for sterile insect technique <sup>5</sup>, in part because of the expense of such facilities <sup>8,62</sup>. For many SIT applications, X-ray is perceived as a more practical machine-generated isotope alternative compared to eBeam <sup>62,63</sup>. However, for *A. transitella*, the combination of a high throughput requirement and a high sterilizing dose (300 Gy) suggests that eBeam might be a more favorable gamma replacement than X-ray. Electron beam categories include medium energy (300 keV to 5 MeV) and high energy (> 5 MeV) <sup>64</sup>. The shielding for medium energy systems can be self-contained within the unit, whereas high energy systems generally need to be housed in concrete bunkers for sufficient shielding <sup>64</sup>. Previously, high energy systems have been considered as a requirement for the penetration required for sterile insect technique applications <sup>8</sup>, but there are ongoing attempts to adapt medium energy eBeam systems for use with SIT <sup>13,63</sup>.

## 5 Conclusion

The results presented in this study show that it is possible to use a commercially available electron beam facility to successfully sterilize *A. transitella*. Electron beam technology can deliver a higher throughput without negatively affecting *A. transitella* compared to currently available X-ray

irradiators. Based on these results, electron beam technology may be a viable alternative to X-ray sterilization, when considering options for the replacement of gamma irradiators in SIT programs.

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Mention of trade names or commercial products in this publication is solely for the purpose of providing specific information and does not imply recommendation or endorsement by the U.S. Department of Agriculture or the Agricultural Research Service. USDA is an equal opportunity provider and employer.

## 7 Conflict of Interest Declaration:

The authors declare no conflicts of interest

## 8 Author contribution statement

- Conceptualization – LL, CB, HW
- Data curation – LL
- Formal analysis – LL, CB
- Funding acquisition – HW, CB
- Investigation – LL, CB
- Methodology – LL, CB, HW
- Project administration – CB, HW
- Resources – CB, HW
- Supervision – CB, HW
- Validation – LL, CB
- Visualization – LL
- Writing – original draft – LL, CB
- Writing – review & editing – LL, HW, CB

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# 10 Tables

**Table 1** Mean measured dose( $\pm$ SE) in Gy of six alanine tablets per intended dose, run and experiment and how the doses are labeled in the following figures to enhance visual clarity. Both lab controls and trip controls were not exposed to irradiation, therefore lab controls of first runs (\*) are marked with green crosses instead of x's, lab controls of second runs (\*\*) are depicted as purple squares instead of purple rings. For lab controls, no alanine dose readings were conducted. For run 2 of the fertility experiment (\*\*\*), due to lack of emerging moths, no lab control could be included in the experiment.

| Intended Dose | Fitness            |                 |                    |       | Fertility          |                 |                    |       |
|---------------|--------------------|-----------------|--------------------|-------|--------------------|-----------------|--------------------|-------|
|               | Measured Dose (Gy) |                 | Depicted Dose (Gy) |       | Measured Dose (Gy) |                 | Depicted Dose (Gy) |       |
|               | Run 1              | Run 2           | Run 1              | Run 2 | Run 1              | Run 2           | Run 1              | Run 2 |
| Lab Control   | -                  | -               | 0*                 | 0**   | 0                  | -***            | 0*                 | -     |
| Trip Control  | 0                  | 0               | 0                  | 0     | 0                  | 0               | 0                  | 0     |
| 200 Gy        | 134.0 $\pm$ 1.9    | 171.3 $\pm$ 1.1 | 134                | 171   | 177.3 $\pm$ 0.7    | 176.4 $\pm$ 0.5 | 177                | 176   |
| 300 Gy        | 175.0 $\pm$ 0.8    | 205.7 $\pm$ 1.1 | 175                | 206   | 210.7 $\pm$ 0.9    | 208.0 $\pm$ 0.6 | 211                | 208   |
| 400 Gy        | 302.0 $\pm$ 0.8    | 323.7 $\pm$ 5.8 | 302                | 324   | 405.0 $\pm$ 0.3    | 310.0 $\pm$ 1.2 | 405                | 311   |
| 500 Gy        | 376.0 $\pm$ 3.0    | 520.7 $\pm$ 3.6 | 376                | 521   | 445.5 $\pm$ 15.0   | 309.5 $\pm$ 1.7 | 446                | 310   |

**Table 2** Analysis of deviance table, showing type II likelihood ratio chi-square tests of main effects and all two-way interactions for the proportion of adults in experiment 1 that escaped from the flight cylinders (GLM with quasibinomial distribution, residual df = 65).

| Effect            | $\chi^2$ | df | P       |
|-------------------|----------|----|---------|
| Dose (Gy)         | 0.001    | 1  | 0.98    |
| Sex               | 43.83    | 1  | < 0.001 |
| Run (Cohort)      | 2.38     | 1  | 0.12    |
| Dose $\times$ Sex | 0.061    | 1  | 0.80    |
| Dose $\times$ Run | 0.129    | 1  | 0.72    |
| Sex $\times$ Run  | 0.018    | 1  | 0.89    |

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**Table 3** Analysis of deviance table, showing type II likelihood ratio chi-square tests of main effects and all two-way interactions for the proportion of adults in experiment 1 that survived >14 days (GLM with quasibinomial distribution, residual df = 65).

| Effect       | $\chi^2$ | df | P       |
|--------------|----------|----|---------|
| Dose (Gy)    | 0.650    | 1  | 0.42    |
| Sex          | 5.056    | 1  | < 0.025 |
| Run (Cohort) | 3.761    | 1  | 0.052   |
| Dose × Sex   | 3.870    | 1  | 0.049   |
| Dose × Run   | 2.439    | 1  | 0.118   |
| Sex × Run    | 0.674    | 1  | 0.412   |

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**Table 4** Analysis of deviance table, showing type II likelihood ratio chi-square tests of main effects and all two-way interactions for the proportion of adults in experiment 2 that mated (GLM with binomial distribution, residual df = 653).

| Effect       | $\chi^2$ | df | P     |
|--------------|----------|----|-------|
| Dose (Gy)    | 0.150    | 1  | 0.698 |
| Sex          | 2.996    | 1  | 0.083 |
| Run (Cohort) | 0.576    | 1  | 0.448 |
| Dose × Sex   | 0.271    | 1  | 0.602 |
| Dose × Run   | 0.144    | 1  | 0.704 |
| Sex × Run    | 0.988    | 1  | 0.320 |

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**Table 5** Analysis of deviance table, showing type II likelihood ratio chi-square tests of main effects and all two-way interactions for the total number of eggs laid by treated adults in experiment 2 (GLM with negative binomial distribution, dose categorical, residual df = 333).

| Effect       | $\chi^2$ | df | P       |
|--------------|----------|----|---------|
| Dose (Gy)    | 23.5     | 1  | < 0.001 |
| Sex          | 2.57     | 1  | 0.12    |
| Run (Cohort) | 0.92     | 1  | 0.34    |
| Dose × Sex   | 5.73     | 5  | 0.33    |
| Dose × Run   | 0.59     | 2  | 0.74    |
| Sex × Run    | 21.3     | 1  | < 0.001 |

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## 579 11 Figure Legends

580 **Figure 1** Effect of eBeam irradiation indicated by dose (Gy) on flight ability of female (red rings)  
581 and male (cyan x's) *A. transitella*. Symbols indicate individual flight cylinder experiments (groups  
582 of 10 moths). The lines indicate the linear regression fit for each sex, and the grey area  
583 indicated the 95% prediction interval.

584 **Figure 2** Effect of eBeam irradiation indicated by dose (Gy) on the proportion of *A. transitella* that  
585 survived >14 d. The lines indicate the linear regression fit for each sex, and the grey area indicates  
586 the 95% prediction interval.

587 **Figure 3** Effect of eBeam irradiation indicated by dose (Gy) on the mating ability of female and  
588 male *A. transitella* indicated by the proportion of mated individuals after two days in the first (red  
589 rings) and second (cyan x's) runs with an untreated individual of the opposite sex out of groups  
590 of ten. Linear regression and the 95% confidence interval for each level of sex × run are  
591 respectively indicated by the straight lines and shaded areas.

592 **Figure 4** Box plots illustrating the effect of eBeam irradiation on the fecundity of *A. transitella*. Sex  
593 indicates the sex irradiated; for “Male” this means eggs per non-irradiated female paired with an  
594 irradiated male. Treatment groups in the same panel with different letters are significantly  
595 different. The 0 treatment indicates moths that were not taken to the eBeam facility, whereas the  
596 5 Gy treatment indicates moths that were taken to the eBeam facility but not exposed.

597 **Figure 5** Box plots indicating pupae per female following successful mating. Treatment groups in  
598 the same panel with different letters are significantly different (treatments with 0 pupae were not  
599 included in this analysis). The numerals indicate the percent of pairings that produced >0 pupae.

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## 601 12 Data availability statement

602 Datasets and analysis scripts used in this study are available from the authors on reasonable  
603 request.

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